

5.7.9

AI25BTECH11036-SNEHAMRUDULA

Question:

$$A = \begin{bmatrix} -3 & 6 \\ -2 & 4 \end{bmatrix},$$

show that $A^3 = A$.

Given: $A = \begin{pmatrix} -3 & 6 \\ -2 & 4 \end{pmatrix}$. The characteristic equation is

$$f(\lambda) = |A - \lambda I| = \begin{vmatrix} -3 - \lambda & 6 \\ -2 & 4 - \lambda \end{vmatrix} = (-3 - \lambda)(4 - \lambda) - (-12) = \lambda^2 - \lambda = \lambda(\lambda - 1) = 0. \quad (1)$$

By CayleyâHamilton, $f(A) = 0$, hence

$$A^2 - A = 0 \implies A^2 = A. \quad (2)$$

Multiplying (2) on the right by A yields

$$A^3 = A^2 A = A A = A^2. \quad (3)$$

Using (2) in (3),

$$A^3 = A. \quad (4)$$

Action of A on vectors (Blue = original, Red = $A \cdot v$)

